" If You Want To Manage It, Measure It "

IOTAFLOW
SYSTEMS PVT. LTD

RPD SERIES FLOW METER

www.iotaflow.com

OIL FLOW METERS FOR MOST DEMANDING APPLICATIONS

Accurate fuel oil receipts. Fuel consumption monitoring in industrial power generators, boilers, burners. Flow rate monitoring & control in close loop systems for bearing lubrication etc.

Features

1. The complete range of products offering the best solutions for the measurement of oil consumption.
2. State-of-the-art design with electronic counter, flow indication, analogue and digital output signals and limiting value switch
3. Mounting on the pressure or suction side of a pump, with no straight inlets or outlets required

4. Independent of viscosity and temperature
5. High vibration resistance
6. Programme-able unit of measurement - Ltrs/m³/gallon

Your benefits

- The reliable solution with everything from a single supplier
- Reliable monitoring and flexible control of the system.
- Simplifies burner settings and optimising consumption
- Highly flexible mounting with very small space requirements
- Accurate measurements
- Maximum safety in the shipbuilding and automobile industries
- Cost-effective metering point

Metoil Electronic Display Flow Meter with multi functional display and parameterisable outputs DN 15... 80

- Electronic display of totaliser, total and resettable volume actual flow rate other flow parameters
- Output signals for volume pulses actual flow rate limiting values (Qmin, Qmax)
- Simple to operate
- Interactive parameter input

Signal

- Analog 4 – 20 mA continuous for remote monitoring and control Digital Pulses
- External power supply
- Battery Back-up option
- Housing with threaded or flanged connections
- Different Construction Material options for varied applications

Main Characteristic Data:

- Flow range 10 ... 48 000 l/h
- Temperature ranges 130 and 180°C
- Nominal pressure PN 16, 25 and 40 bar

The advantages of Metoil® fuel oil meters – your benefits

You can decide which of these many benefits are the most important for you:
The optimal solution for every application simple burner setting with flow rate display simple consumption monitoring with limiting value switch Qmin/Qmax manual dosing feature, with a resettable counter can be mounted on the pressure or suction side of a pump space-saving installation, because no straight inlet/outlet sections are needed flexible mounting of the meter in horizontal, vertical or inclined positions accurate measurement result, since the reading is independent of the temperature and viscosity of the fluid minimum failure costs due to simple function monitoring, rapid fault analysis and the possibility of simple repairs on site

Areas of Application

To measure heating fuel consumption by oil burners (for example, in heating boilers, industrial furnaces, tar processing plants, ships and boilers) to measure propellant fuel consumption by motors and engines (such as diesel locomotives, construction machinery and ships, or in emergency power units, combined heating and power stations) consumption monitoring and optimisation flow measurement for mineral oils optional remote processing and integration into superior systems manual dosing / batching flow measurement for machine and motor/engine oils engine test benches

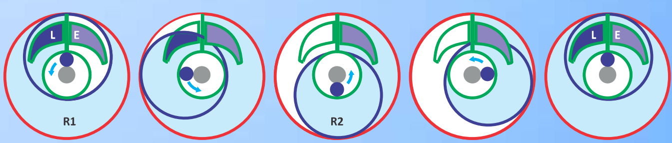
Fuel Types

Heating fuel extra light / Light, Medium,
Heavy naphtha diesel petrol and
other Lubricating liquids

Technical data

Function

Metoil® flow meters work on the volumetric principle of rotary piston meters (positive displacement meters).



The Ring Piston carries out a rotational movement within the flow meter chamber. due to the eccentric movement of the piston, the space within the ring & outer space between the wall of the flow meter is filled through the inlet opening **L** after the filling sequence, the fuel is emptied through the outlet opening **E**. The movement of the ring piston is recorded by means of electronic sensors. One Rotation of the flowmeter corresponds to a defined flow rate.

The main features of this measuring principle are large measuring ranges, high accuracy, suitability for high viscosities and independence from power supply; flow disturbances do not influence proper operation.

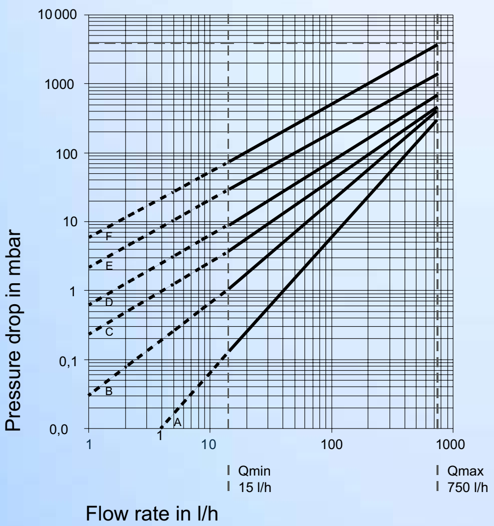
Construction

Piston, roller and driver are the only moving parts in contact with the liquid. Their movement is transmitted by a magnetic coupling through a sealing plate. The hydraulic part is completely separated from the totalising module.

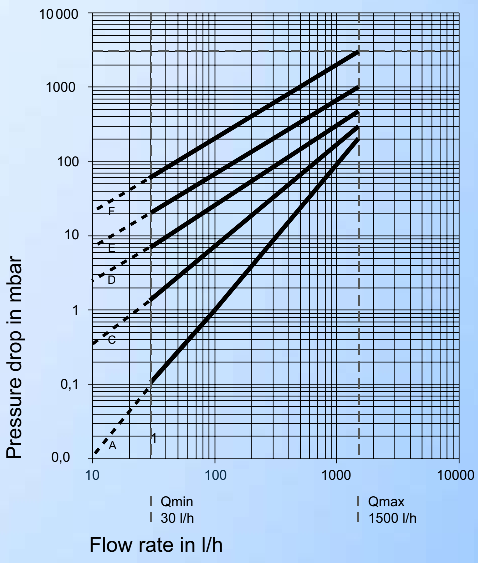
Nominal Diameter	Unit	DN 4	DN 8	DN 15	DN 20	DN 25	DN 40	DN50	DN80
Min. Flowrate	LPH	1	4	15	30	100	300	600	1200
Normal Flowrate	LPH	50	135	500	1000	2000	6000	12000	24000
Max. Flowrate	LPH	80	200	750	1500	4000	12000	24000	48000
Min. counter registration	Ltrs / m^3	0.1	0.1	0.1	0.1	0.1	1	1	1
Indicating range	Ltrs / m^3	999999.9	999999.9	999999.9	999999.9	999999.9	999999.9	999999.9	999999.9
Temperature	deg C	60	60	130/180	130/80	130/80	130/80	130/80	130/80
Nominal Pressure	bar	25/32	25	16/25/40	16/25/40	16/25/40	25/40	25/40	25/40
Connection		Scr ewed	Scr ewed	Scr ewed / Flanged	Scr ewed / Flanged	Scr ewed / Flanged	Flanged	Flanged	Flanged

Pressure drop curves

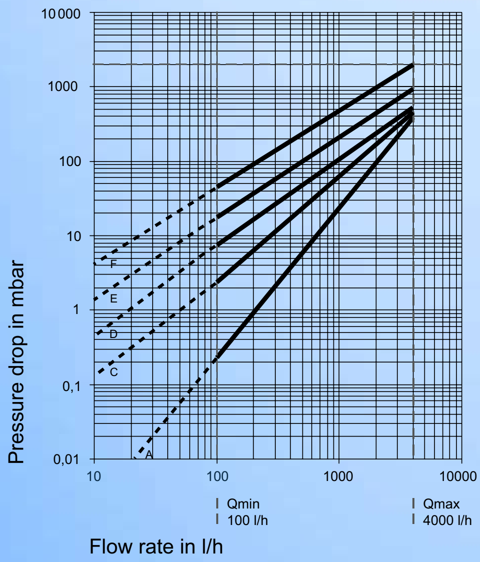
DN 15



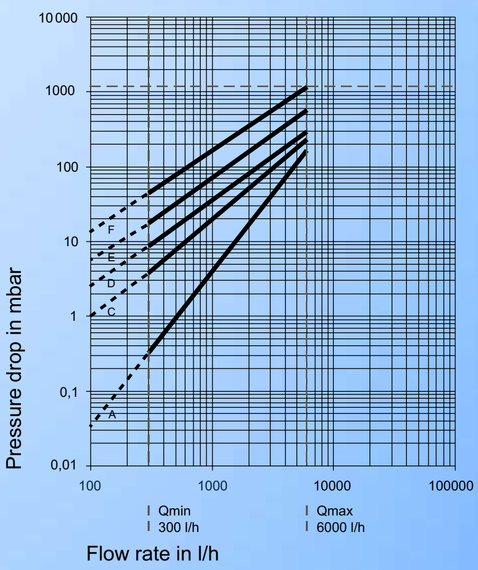
DN 20



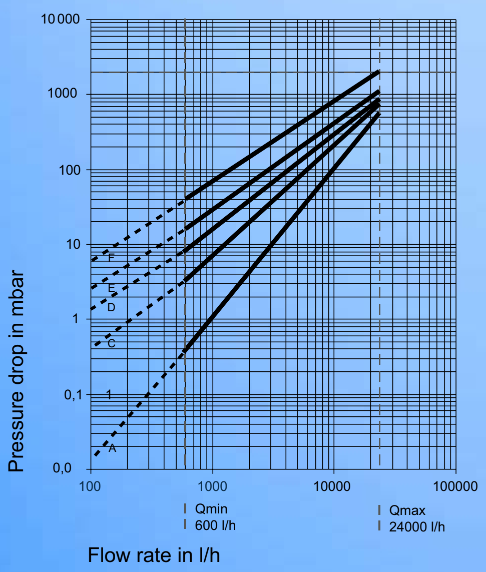
DN 25



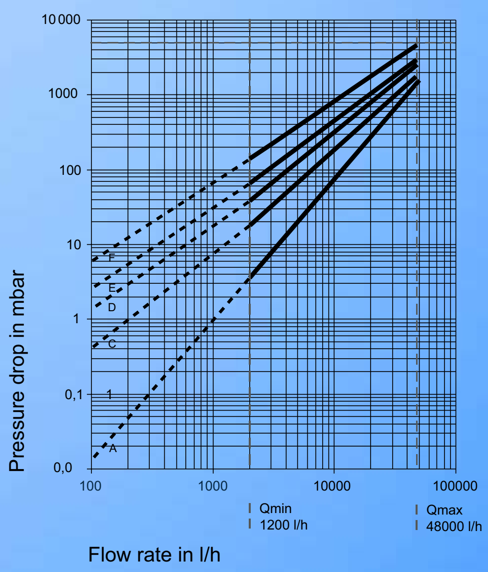
DN 40



DN 50



DN 80



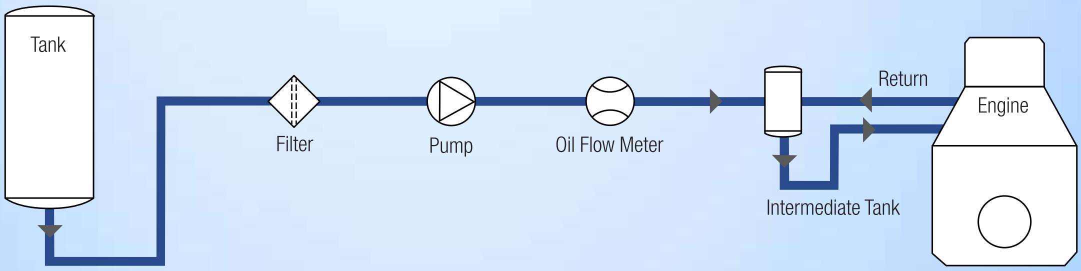
Viscosity diagrams:
A = 5 mPas, B = 25 mPas, C = 50 mPas, D = 100 mPas, E = 200 mPas, F = 500 mPas
For a pressure drop of more than 1 bar, it is recommended to use the next larger meter size

Application Examples

Diesel engine

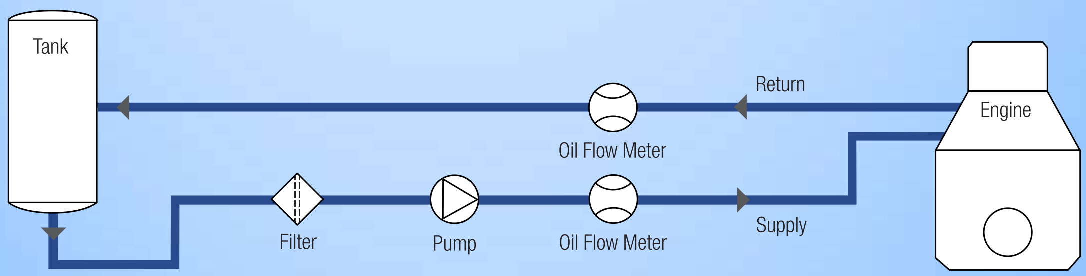
Direct consumption measurement

Instead of returning the fuel back into the main tank, an intermediate tank equipped with a heat exchanger should be installed on the supply side of the system. The flow measurement is taken in the supply pipe to the intermediate tank. The load on the meter and the measuring result correspond precisely to the consumption.



Differential measurements

For differential measurements, the piping remains unchanged, with circulation back into the tank. A flowmeter is installed in both supplyand return pipes. The consumption is determined as the difference between the amount in the supply section and the amount in the return section. The meter loads therefore correspond to the supply and return flow rates.



Reasons for using special meters for differential measurements

Standard meters feature a large measuring range and a max. permissible error of ±1%. This makes them unsuitable for differential measurements, as the following example shows:

Full load	Supply	400 l/h	Error ±1%	= nominal ±4.0 l
	Return	150 l/h	Error ±1%	= nominal ±1.5 l
	Consumed	250 l/h	Divergence	nominal ±5.5 l

Maximum divergence
Consumed = 5.5 100 : 250 = ±2.2 %

Min. load	Supply	400 l/h	Error ±1%	= nominal ±4.0 l
	Return	360 l/h	Error ±1%	= nominal ±3.6 l
	Consumed	40 l/h	Divergence	nominal ±7.6 l

Maximum divergence
Consumed = 7.6x100 : 40 = ±19 %

For an optimal result, special meters are therefore used for differential measurements. These are precisely matched to the operating conditions and are calibrated in pairs. This means that the measurement error can be significantly reduced (for example: ±0.1% at constant flow rates on the supply side and ±0.3% with slightly variable flow rates on the return side)

Following options make the selection of the flow meter very easy & at the same time versatile

RPD XX X X X X X X X X X X X X

Size	DN 4 to DN 80	XX
Display Type	Mechanical	M
	Digital Flow Rate only	F
	Digital Flow Rate & Total	D
Body Material	Brass	B
	Cast Iron	C
	Cast Steel	D
	SS 304	S
	SS 316	F
Measuring Chamber Material	Brass	B
	SS 304	S
	SS 316	F
	PTFE	P
Piston Material	Anodized Aluminium	A
	SS 304	S
	SS 316	F
	PTFE	P
	Hard Rubber	H
	Graphite	G
Temp. Suitabilty	60 °C	60
	130 °C	130
	180 °C	180
Accuracy	1.0%	1
	0.5%	0.5
Seals	FPM	F
	PTFE	P
	Viton	V
	FFKM	K
Output	No output	O
	4-20 mA	A
	Pulse	D
Input	5 VDC	L
	10 - 30 VDC	V
	Battery	A
Pressure	16 bar	16 B
	25 bar	25 B
	32 bar	32 B
	40 bar	40 B
End Connection	S crewed	S
	Flanged PN 16	PN 16
	Flanged PN 25	PN 25
	Flanged PN 40	PN 40
	Flanged # 150	150
	Flanged # 300	300
Display Enclosure	IP 63	A
	IP 65	B
	Ex-Proof	E
	IP 66	C

Contact Us

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